RMIT MASTERS THESIS — TECHNICAL DEEP DIVE

Following the Money: From Transactions to Criminal Networks

Detection, Ranking and Visualisation of Money Laundering Networks on the Bitcoin Blockchain

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Detection, Ranking and Visualisation of Money Laundering Networks on the Bitcoin Blockchain

The goal of this thesis was simple: **follow the money**. Not just flag suspicious transactions, but trace illicit funds upward through a criminal network — from low-level money mules to the most influential actors in the operation.

The aim is to give investigators the intelligence and tools to **take down the entire criminal organisation quickly and efficiently**, not just pick off individuals at the edges.

The result is a transparent, explainable, cloud-deployed system that an investigator can use and a regulator can audit.

> **THIS PRESENTATION**
> A technical deep dive into the methodology, algorithms, and architecture behind each component of the thesis.

MOTIVATION

Why I Built This

From research and experience, I made four observations about the state of AML detection:

Crime is evolving faster than detection

Money laundering is moving beyond traditional banking into cryptocurrencies, cross-chain swaps, mixers, and decentralised exchanges — exploiting pseudonymity, speed, and global reach. Almost USD 100 billion in illicit crypto was transferred through conversion services between 2019 and 2024

Full-graph methods aren't realistic

Current network solutions analyse entire transaction graphs accumulated over months or years. In practice, detection is time-sensitive — investigators have only a narrow window of recent data and can't wait for a complete picture

Black boxes fail regulation

Pushing innovation that produces black-box solutions will fail regulatory scrutiny — every model choice must be explainable to regulators

Not built for real life

Solutions remain exploratory or academic — stand-alone models with no clear path to implementation in a real financial crime department. They ignore time constraints, investigative workflows, and operational realities

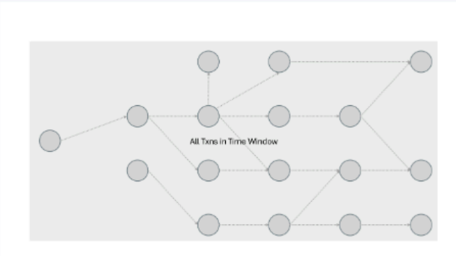
THE GOAL

Build an investigator-focused, regulator-explainable solution with the potential to capture criminal networks — not just suspicious individuals. If it doesn't work for investigators, it won't be implemented.

METHODOLOGY

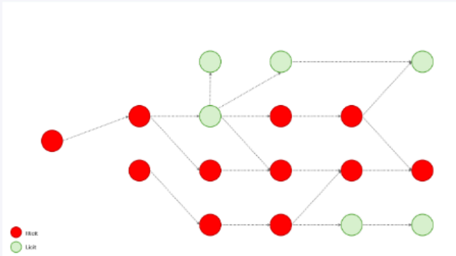
A Multi-Layered Approach

1. CLASSIFY	→	2. BUILD NETWORKS	→	3. RANK	→	4. VISUALISE
Supervised classification of transactions		Graph-based subnetwork construction		Composite ranking of key entities		Interactive investigator dashboard



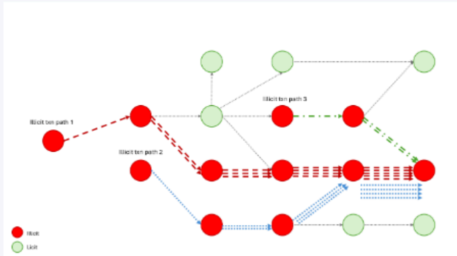
STEP 1

Get transactions for a time window with features and edge lists showing BTC flow



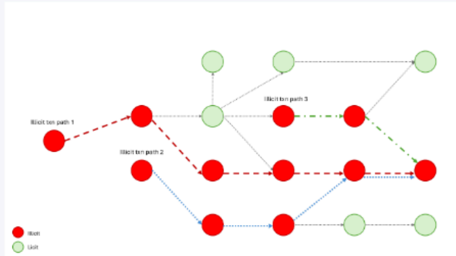
STEP 2

Classify each transaction as illicit or licit using Random Forest



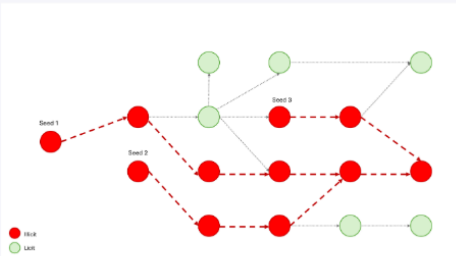
STEP 3

Build subnetworks via forward BFS from each illicit seed, stopping at licit boundaries



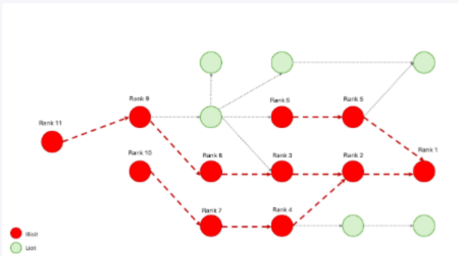
STEP 4

Deduplicate subnetworks with 100% overlap of transactions



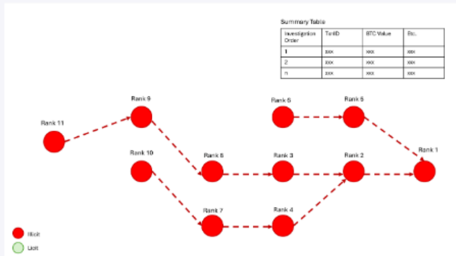
STEP 5

Merge subnetworks sharing at least 1 illicit node into a single network



STEP 6

Rank each transaction by network importance (PageRank) + BTC value



STEP 7

Visualise ranked network + summary table for investigators

BACKGROUND

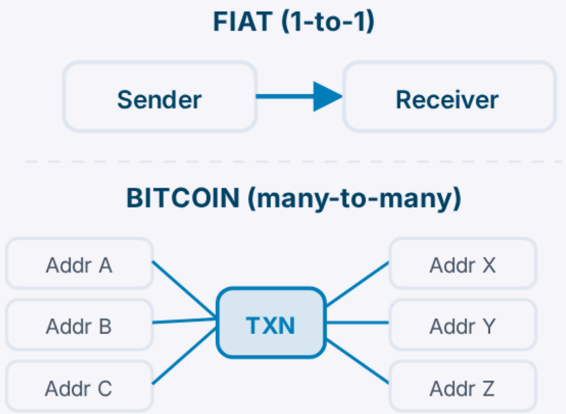
Bitcoin & the Blockchain

How the Blockchain Works

Transactions grouped into blocks, cryptographically chained. Public, immutable, decentralised — no intermediary.

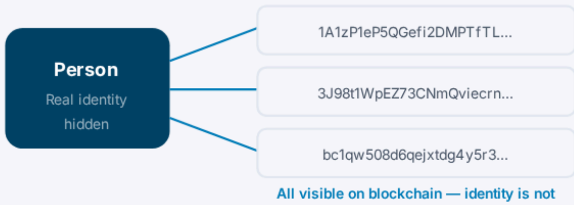
Fiat vs Bitcoin Transactions

Traditional banking is 1-to-1. Bitcoin's UTXO model is many-to-many — fundamentally different for AML.



Pseudonymity

Users generate unlimited addresses. Visible on the blockchain, but not tied to real-world identity.



Why This Matters for AML

The public ledger is a double-edged sword — transparency exists but so do obfuscation strategies (mixing, chain-hopping, cross-chain swaps). Traditional AML systems aren't designed for many-to-many transaction structures. ~USD 100B in illicit crypto transferred via conversion services between 2019–2024.

TAKEAWAY

Bitcoin's pseudonymous, many-to-many transaction model creates both an opportunity (public ledger) and a challenge (obfuscation) for financial crime detection.

See Appendix: AI Quick Guide

DATASET

The Elliptic++ Dataset

203,769	2%	21%	77%	182
Transactions	Illicit (4,545)	Licit (42,019)	Unknown (157,205)	Features

This thesis uses the **Elliptic++ dataset** (Elmougy & Liu, 2023), which extends the original Elliptic dataset — Bitcoin transaction data from a regulated blockchain analytics firm spanning ~2016–2017 across 49 time steps. The dataset includes transaction features, transaction-to-transaction edges, wallet features, and address-to-transaction / transaction-to-address mapping tables.

Features

93 local — transaction-level (inputs, outputs, fees, volume)

72 aggregate — one-hop neighbourhood statistics

17 unmasked (E++) — BTC flows, address counts, txn size

822K+ wallet addresses via deanonymisation

Supporting: txn-txn edges, wallet features, addr-txn / txn-addr mappings

Constraints

Severe class imbalance — only 2% illicit

Proprietary labels — can't be verified

No edge-level BTC weights

No wallet-type metadata

Validation

Spot checks on deanonymised addresses using 3rd party blockchain explorers (BTC Scan)

Confirmed **structural consistency** of transaction metadata

Labels could not be independently verified (proprietary source)

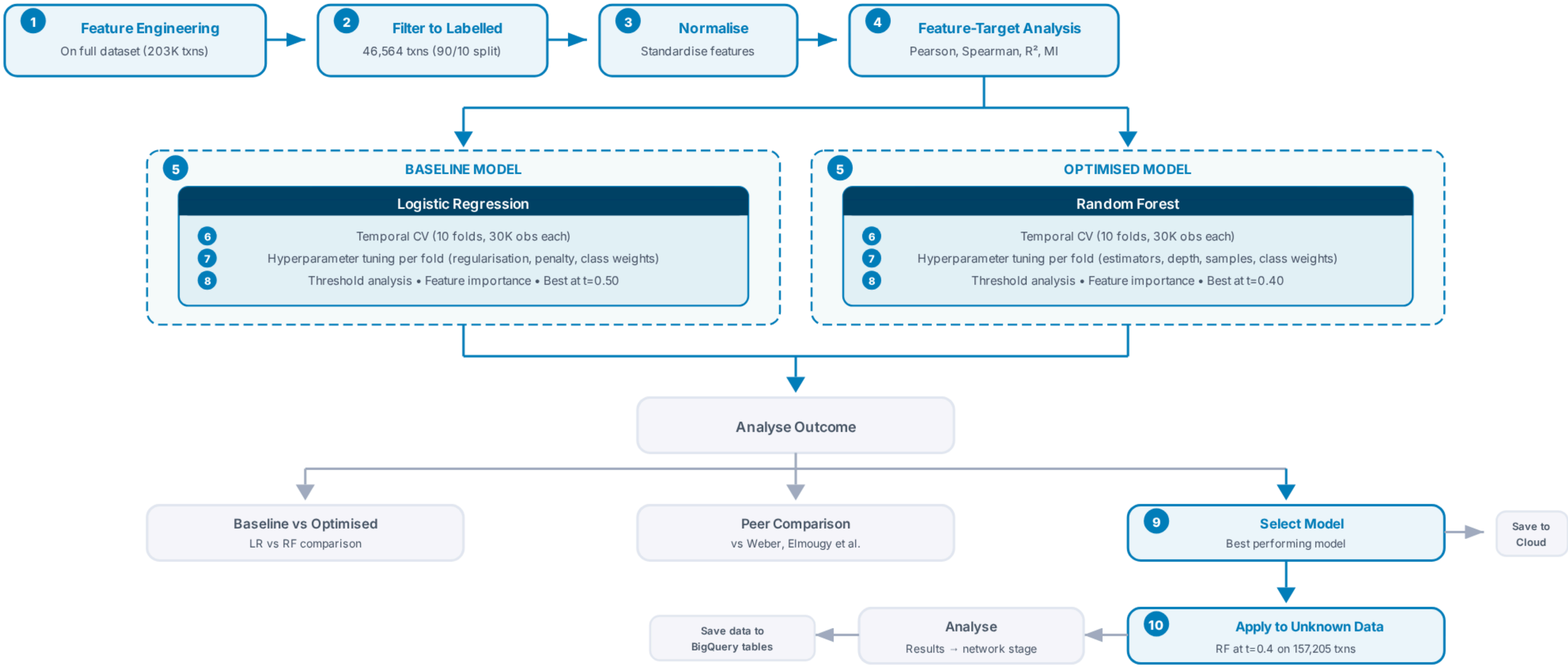
TAKEAWAY

Elliptic/Elliptic++ is the largest and most credible labelled Bitcoin dataset publicly available — developed by a regulated analytics firm, not scraped from community sources.

CLASSIFICATION

Classification Training

The classification training process was built in Jupyter Notebook — iterating through feature engineering, model training, and validation before selecting a final model for production. Every step was designed to prevent data leakage and falsely inflated performance. The following slides walk through each stage.



Classification Training Pipeline

The classification pipeline was designed to prevent data leakage, ensure reproducibility, and produce results that hold up under regulatory scrutiny. Every step has a methodological justification.

#	STEP	WHY	RISK IF DONE POORLY
1	**Feature Engineering**	The existing dataset features describe general transaction behaviour. Additional features can be engineered from domain knowledge to capture laundering-specific patterns the original data wasn't designed to detect.	The model only learns generic transaction patterns — can't distinguish laundering from normal activity.
2	**Filter to Labelled**	Classification requires labelled data. The 77% unknown transactions were filtered out for training — they return later for prediction.	Cannot verify model performance if training on unknown labels.
3	**Normalise**	Features vary in scale (BTC values in thousands, flags as 0/1). Critical for LR where coefficients are scale-dependent; less impactful for RF which splits on rank order, but ensures consistency across models.	LR coefficients become meaningless — the model learns scale, not signal. Cross-model comparisons unreliable.
4	**Feature-Target Analysis**	195 features is a lot. The shape of the data determines which model architecture is appropriate — linear, non-linear, or ensemble.	Selecting an architecture that mismatches the data — e.g. a linear model for non-linear data. No empirical basis for selection.
5	**Model Choice**	LR is a well-established baseline used across AML literature for decades. RF is a robust, auditable ensemble proven with non-linear data across multiple peer-reviewed studies. Without a baseline, you can't prove the optimised model is worth the complexity.	Poor model selection leads to either missed illicit activity or excessive false alerts — both undermine investigative effectiveness.
6	**Temporal Cross-Validation**	Random splits leak future data into training, falsely inflating performance. Temporal folds ensure the model only sees the past — reflecting real-world deployment conditions.	Reported metrics are falsely high. Model appears strong in testing but fails on genuinely unseen data.
7	**Hyperparameter Tuning**	Hyperparameter tuning performed within each fold — ensuring tuning decisions are free of leakage and independently validated across time periods.	Tuning on leaked data produces overfit hyperparameters that don't generalise to new data.
8	**Threshold Analysis**	For each model, thresholds analysed based on accuracy, precision, recall, F1, and false positive rates. AML requires recall-optimised thresholds.	Drowns investigators in false alerts, or silently misses criminal activity.
9	**Select Model**	Head-to-head comparison + benchmark against peer-reviewed literature.	No defensible justification for the chosen model.
10	**Apply to Unknown**	Selected model on 157,205 unlabelled transactions. Feeds into network stage.	77% of transactions remain unclassified — insufficient network coverage.

CLASSIFICATION

Feature Engineering

166 masked features from Elliptic, 17 unmasked features from Elliptic++, plus 12 additional features engineered from research and domain knowledge to capture laundering-specific behaviour. Total: 195.

Masked Features (Elliptic)

166 features

94 local features — scaled and normalised. Includes time step, number of inputs/outputs, transaction fee, output volume, average BTC received/spent by inputs/outputs, average incoming/outgoing transactions

72 aggregate features — one-hop neighbouring transactions. Max, min, std dev, correlation coefficients of neighbour data (inputs/outputs, fees, etc.)

Refs: Elliptic Blog · Kaggle · Weber et al. 2019

Unmasked Features (Elliptic++)

17 features

Structural: in_txs_degree, out_txs_degree, num_input_addresses, num_output_addresses, size

Financial: total_BTC, fees, in_BTC (min/max/mean/median/total), out_BTC (min/max/mean/median/total)

Refs: GitHub · Youssef et al. KDD 2023

Engineered Features (this thesis)

12 features

in_out_ratio — funds accumulating vs moving through

input_output_ratio — consolidation vs distribution

BTC_dispersion — mixing patterns (max-min output spread)

high_fees_flag — urgency (>95th percentile fees)

micro_txn_flag — dust attacks / structuring (<0.001 BTC)

txn_density — transaction complexity per address

fees_per_byte, fees_ratio, fees_per_input — fee efficiency

rounded_amount_flag — human-initiated structuring

input/output_address_percentile — relative position in dataset

TAKEAWAY

Local features describe *what* a transaction is. Aggregate features describe *who it transacts with* — encoding the one-hop network neighbourhood. Engineered features describe *whether it behaves like laundering*. All 195 features work together, from individual transaction to network context to criminal typology. Laundering is a pattern across transactions, not a property of a single one.

CLASSIFICATION

Feature-Target Analysis

Before selecting a model, understand the shape of the data. Not all features contribute equally — the type of relationship (linear vs non-linear) determines which models can capture the signal. If the data is fundamentally non-linear, a linear model is structurally incapable of learning the pattern.

Four tests on each of 195 features

- **Pearson** — linear relationship strength
- **Spearman** — monotonic, captures non-linear patterns
- **R^2** — variance explained by a linear fit
- **Mutual information** — any statistical dependency

Feature Categorisation Rules

Based on established statistical conventions for identifying meaningful relationships.

- **Linear:** $|\text{Pearson}| > 0.5$ AND $R^2 > 0.3$
- **Non-linear:** $|\text{Pearson}| < 0.3$ BUT $|\text{Spearman}| > 0.5$ OR MI > 0.1
- **Weak:** neither criterion met

Results

- **Zero features** have a linear relationship with illicit activity
- **42 features** carry signal through non-linear interactions only
- **153 features** are individually weak predictors
- RF is designed to aggregate weak, non-linear signals — architecturally matched to this data

A glimpse of the per-feature results:

	feature	pearson	spearman	r2	mutual_info	relation
2	Local_feature_1	0.0326	0.0747	0.0011	0.0353	weak
3	Local_feature_2	-0.0740	-0.1226	0.0055	0.1597	nonlinear
4	Local_feature_3	-0.0804	-0.0903	0.0065	0.0113	weak

TAKEAWAY

The feature-target analysis reveals the shape of the data — fundamentally non-linear with no linear features. This doesn't pick a model, but it narrows the field: models that can aggregate weak, non-linear signals are likely better suited than those that assume linearity.

CLASSIFICATION

Model Selection Rationale

Two models chosen deliberately — a transparent baseline and an optimised classifier. Both are interpretable, auditable, and deployable in a regulated environment.

Models Chosen

Logistic Regression (Baseline)

- Transparent baseline widely used in AML research (Jullum et al., Lorenz et al.)
- Every feature gets a single coefficient — fully explainable to regulators and auditors
- Establishes whether detectable signal exists before introducing model complexity

Random Forest (Optimised)

- Weber et al. tested LR, RF, MLP, and GCN on Elliptic — RF outperformed all others
- Consistently outperformed deep/complex neural networks on imbalanced AML datasets across multiple peer-reviewed studies
- Feature-target analysis: **42 non-linear, 0 linear** relationships across 195 features — empirical justification for tree-based over linear models
- Interpretable via feature importance — every prediction traceable through individual trees

Models Considered & Excluded

XGBoost / Gradient Boosting

Sequential boosting makes individual predictions harder to audit. Elmougy's RF+MLP+XGB ensemble added complexity without meaningful gain over standalone RF on this dataset.

Graph Convolutional Networks (GCNs)

Underperformed RF in Weber et al. Black-box architecture — predictions cannot be explained at the individual transaction level. Excluded to maintain transparency and auditability.

Deep Learning / MLPs

Weber et al. found MLP did not outperform RF. No feature-level explainability. Excluded for regulatory transparency.

Support Vector Machines

Common in early AML literature. No prior work applied SVMs to Elliptic for comparison. Scale poorly with high-dimensional, imbalanced data.

TAKEAWAY

Deep learning, GNNs, and complex ensembles were excluded deliberately. In a regulated AML environment, every prediction must be explainable to an auditor. LR and RF are transparent, well-precedented, and empirically validated on this data.

See Appendix: Traditional ML Guide · Classification Models Overview

Temporal Cross-Validation

Bitcoin transactions are inherently time-ordered, so validation must respect that temporal structure. Each fold samples **30,000 consecutive labelled observations (~64% of the labelled population)**, splits them 50/50 by time step, and tunes hyperparameters independently on the training portion. Across 10 folds, both LR and RF are evaluated on different temporal windows — ensuring consistent, leak-free performance before threshold analysis selects the final operating point.

How each fold works

1. **Sample 30K consecutive observations**
   From 46,564 labelled transactions

2. **Split 50/50 by time step** Train Val
   15K earlier = Train • 15K later = Validation

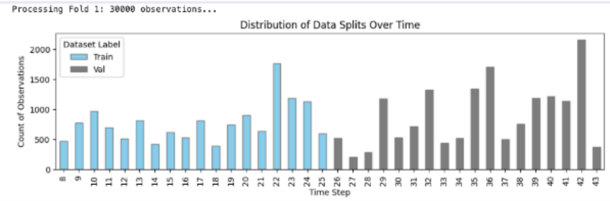
3. **Hyperparameter tuning (train only)**
   HalvingRandomSearchCV • 3-fold inner TimeSeriesSplit

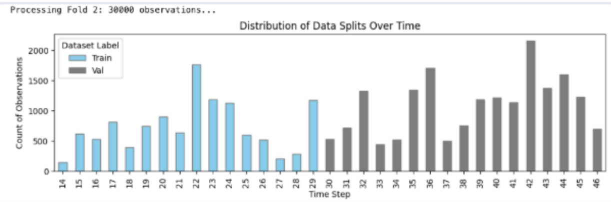
4. **Evaluate on held-out test portion**
   Accuracy, Recall, Precision, F1, AUC-ROC, Gini

This process repeats **10 times** — each fold slides forward in time. The model is never tested on data it has seen.

Fold examples — sliding time window

Each fold selects 30,000 observations and splits by time step. Blue = train, grey = validation.





TAKEAWAY

10 folds, each with a different temporal window, ensures the model is consistently strong across time — not just on one favourable period. Prior work on Elliptic used single train/validation splits; this temporal rigour is the primary reason this study outperforms them.

Hyperparameter Tuning

Hyperparameters are settings chosen *before* training that control how the model learns — they can't be learned from the data. Each model was first trained with baseline settings, then re-trained with systematic optimisation across all 10 folds. **40 models** independently trained (4 runs × 10 folds: 10 LR baseline, 10 LR tuned, 10 RF baseline, 10 RF tuned).

LR: *"Best way to draw a straight line?"*

PARAMETER	BASELINE	TUNED
Regularisation	L2 (default)	L1, L2, Elastic Net
Penalty (C)	1.0 (default)	0.03, 0.1, 0.3, 1.0
l1_ratio	—	0.0, 0.5, 1.0
Class weighting	None	Balanced
Solver	Default	saga
Search method	—	HalvingGridSearchCV

RF: *"How complex, how diverse?"*

PARAMETER	BASELINE	TUNED
n_estimators	200 (fixed)	100 → 1,000
max_depth	None	None, 12–36
min_samples_split	2 (default)	2–20
min_samples_leaf	1 (default)	1–20
max_features	Default	sqrt, log2, None
Class weighting	Balanced	Balanced, subsample
Search method	—	HalvingRandomSearchCV

Both tuned with **recall** as scoring metric • Inner CV: TimeSeriesSplit (3 folds) • Temporal discipline maintained throughout

Before and after tuning (best fold per run • all at t=0.50)

MODEL	RECALL		PREC.		GINI	
LR Baseline	0.835		0.752		0.938	
LR Tuned	**0.945**	↑	0.463	↓	0.910	↓
RF Baseline	0.899		0.997		0.976	
RF Tuned	**0.908**	↑	**0.975**	↓	**0.989**	↑

Key findings

LR: Recall pushed from 83.5% to 94.5% — but precision collapsed to 46.3%. The linear model catches more, but only by flagging far more innocent transactions.

RF: Gini improved (0.976 → 0.989) and recall rose (89.9% → 90.8%) while precision stayed near-perfect (97.5%). Improves without trade-offs.

TAKEAWAY

After tuning, for every 100 transactions LR flags as suspicious, 54 are innocent — investigators waste half their time chasing false leads. RF flags 100 and 97 are genuinely suspicious. Hyperparameter tuning confirmed this isn't a configuration problem — 40 models were systematically explored and LR's precision still collapsed. The limitation is structural: no amount of tuning can make a linear model capture non-linear signal.

CLASSIFICATION

Threshold Analysis

The model outputs a probability (0–1) for each transaction. The threshold determines where to draw the line: above = flagged illicit, below = classified licit. After selecting the best model for each type — ranked by Gini, then recall, then accuracy — the selected model was applied to the **full labelled dataset (46,564 transactions)** to generate predictions. Threshold analysis was then conducted across the full population to determine the optimal operating point.

Random Forest (selected: t=0.40)

T	RECALL	PREC.	FP	FN	ACC.
0.10	99%	24%	14,564	49	69%
0.20	94%	47%	4,892	278	89%
0.30	91%	68%	1,948	401	95%
0.40	**88%**	**86%**	**652**	**535**	**97%**
0.50	85%	95%	192	669	98%
0.60	83%	99%	52	752	98%
0.70	78%	99%	25	985	98%
0.80	54%	100%	12	2,071	96%
0.90	47%	100%	2	2,417	95%

RF at t=0.4: 652 false positives, 535 false negatives. For every 100 alerts, **86 are genuine**. The model catches 88% of illicit activity with manageable investigator workload.

Logistic Regression (selected: t=0.50)

T	RECALL	PREC.	FP	FN	ACC.
0.10	98%	26%	12,490	110	73%
0.20	96%	30%	10,141	170	78%
0.30	95%	33%	8,728	223	81%
0.40	94%	36%	7,625	291	83%
0.50	**92%**	**38%**	**6,700**	**380**	**85%**
0.60	89%	41%	5,827	500	86%
0.70	86%	44%	4,921	652	88%
0.80	80%	49%	3,792	901	90%
0.90	71%	58%	2,302	1,309	92%

LR at t=0.5: 6,700 false positives, 380 false negatives. For every 100 alerts, only **38 are genuine**. Precision never exceeds 58% at any threshold — LR cannot cleanly separate the classes.

TAKEAWAY

Choosing a threshold is a compliance decision, not a technical one. At t=0.4, the RF model surfaces 88% of suspicious activity while keeping false alerts manageable (652). LR cannot achieve this balance at any threshold — confirming the model selection decision. The threshold is tunable to organisational risk appetite.

CLASSIFICATION

Model Selection & Peer Comparison

Random Forest was selected as the classification model. While LR achieved higher recall (91.6% vs 88.2%), this came at the cost of precision collapsing to 38.3% — operationally unviable. RF delivered the strongest overall balance of recall, precision, and discriminatory power, making it the only model suitable for deployment.

88.2%	86.0%	0.871	0.933	0.867
Recall	Precision	F1-Score	AUC-ROC	Gini

LR vs RF (full dataset, selected thresholds)

METRIC	LR (T=0.50)	RF (T=0.40)	
Accuracy	0.848	0.975	↑
Recall	0.916	0.882	↓
Precision	0.383	0.860	↑
F1-Score	0.541	0.871	↑
Gini	0.757	0.867	↑

LR's higher recall comes at 38% precision — for every 100 alerts, 62 are innocent. RF balances both: 86 of every 100 alerts are genuine.

vs published literature (Elliptic/Elliptic++ dataset)

#	MODEL	PREC.	RECALL	F1
1	This Study (RF)	0.860	0.882	0.871
2	Elmougy (RF+MLP+XGB)	0.968	0.729	0.834
3	Elmougy (RF)	0.986	0.727	0.836
4	Elmougy (XGB)	0.793	0.718	0.754
5	Elmougy (RF+XGB)	0.987	0.717	0.826
6	Weber (RF)	0.971	0.675	0.796
7	Weber (MLP)	0.780	0.617	0.689
8	Weber (LR)	0.537	0.528	0.533
9	Weber (GCN)	0.812	0.512	0.628
10	Elmougy (LSTM)	0.709	0.223	0.339

+15.3pp recall over Elmougy's best ensemble • +20.7pp over Weber's RF

TAKEAWAY

#1 recall on Elliptic/Elliptic++ — and this is a single transparent model, not a black-box ensemble or GNN. The improvement comes from methodological rigour (temporal CV, hyperparameter tuning, feature engineering), not model complexity. RF is selected as the classification model for the remainder of the pipeline.

Feature Importance

Feature importance measured using two complementary methods on the selected Random Forest model. Permutation importance measures how much F1 drops when a feature is shuffled; Gini importance measures how much each feature contributes to tree splits.

Top 20 by Permutation Importance (F1)

#	FEATURE	PERM. INDEX	GINI RANK
1	**Local_feature_53**	100.0	2
2	**Local_feature_55**	96.4	3
3	**Aggregate_feature_70**	84.1	33
4	**Aggregate_feature_66**	65.6	65
5	**Aggregate_feature_65**	62.9	60
6	**size** (E++)	58.0	11
7	Local_feature_81	47.6	54
8	Local_feature_2	39.3	18
9	Local_feature_90	34.1	19
10	Local_feature_80	29.9	45
11	Aggregate_feature_51	26.6	44
12	Local_feature_59	20.0	32
13	Local_feature_3	18.8	87
14	Local_feature_61	17.5	39
15	Local_feature_18	15.8	22
16	Local_feature_65	14.7	29
17	Local_feature_76	14.3	50
18	Local_feature_47	13.7	1
19	**output_address_pct** (eng.)	13.3	15
20	Aggregate_feature_61	13.0	100

Top 20 by Gini Importance

#	FEATURE	GINI	PERM. RANK
1	**Local_feature_47**	0.066	18
2	**Local_feature_53**	0.066	1
3	**Local_feature_55**	0.059	2
4	Local_feature_41	0.049	—
5	Local_feature_43	0.044	—
6	Local_feature_23	0.039	—
7	**fees_per_byte** (eng.)	0.036	—
8	Local_feature_49	0.036	—
9	Local_feature_29	0.034	—
10	**fees_per_input** (eng.)	0.030	—
11	**size** (E++)	0.029	6
12	Local_feature_31	0.028	—
13	Local_feature_25	0.025	—
14	Local_feature_52	0.019	—
15	**output_addr_pct** (eng.)	0.018	19
16	Local_feature_5	0.015	—
17	**fees** (E++)	0.015	21
18	Local_feature_2	0.015	8
19	Local_feature_90	0.015	9
20	**num_output_addr** (E++)	0.015	—

(E++) = Elliptic++ unmasked feature • **(eng.)** = Engineered feature (this thesis) • Both local and aggregate features dominate — consistent with Weber et al. Three engineered features appear in the Gini top 20, validating the domain-driven feature engineering.

CLASSIFICATION

Applying to Unknown Transactions

The selected RF model (t=0.40) was applied to the 157,205 previously unlabelled transactions to generate predictions. Existing labels were preserved — only unknown transactions received predicted labels. The result is a fully labelled dataset.

How final labels were assigned

ORIGINAL LABEL	ACTION	FINAL LABEL
Illicit (4,545)	Kept as-is	Illicit
Licit (42,019)	Kept as-is	Licit
Unknown (157,205)	RF prediction at t=0.40	23,997 Illicit • 133,208 Licit

Of the unknown transactions, **15.3% were predicted illicit** and **84.7% were predicted licit**. These are probabilistic predictions — not verified ground truth.

Dataset composition

Before

4,545 Illicit (2%)
42,019 Licit (21%)
157,205 Unknown (77%)

After

28,542 Illicit (14%) — 6.3x expansion
175,227 Licit (86%) — 4.2x expansion
0 Unknown

84% of the final illicit group came from previously unknown transactions. The classification model extended coverage well beyond the original labels — surfacing potentially suspicious activity that would otherwise remain invisible.

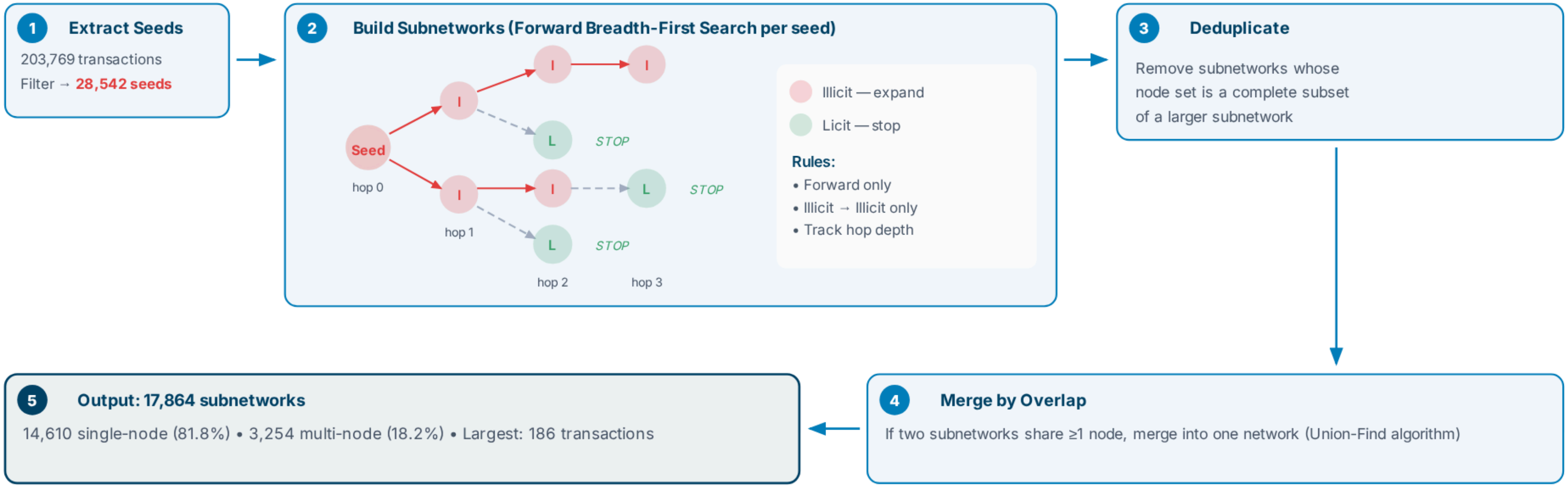
NEXT

A fully labelled dataset is the foundation for network construction. With every transaction now classified and 28,542 potentially illicit transactions identified, the next stage traces how suspicious funds move — following the money from flagged transaction to flagged transaction across the blockchain.

NETWORKS

Network Methodology

Starting from the 28,542 potentially illicit transactions, subnetworks are built by tracing fund flows forward — expanding only through illicit-to-illicit edges and stopping at licit boundaries. By design, the algorithm excludes licit transactions from the network — protecting innocent parties from guilt by association and ensuring investigative focus remains on suspicious activity only.



TAKEAWAY

17,864 subnetworks were constructed — each built deterministically, reproducible, and auditable. Of these, 3,254 multi-node networks (18.2%) represent connected segments of potentially illicit fund flow ready for ranking and investigation. The remaining 14,610 single-node alerts (81.8%) are still flagged as individual suspicious transactions, but without connecting illicit nodes they are treated as standard transaction-level alerts rather than network cases.

See Appendix: BFS Explainer

Network Design Rationale

A classified transaction alone isn't actionable — investigators need to see the network around it. The network construction was designed to bridge classification output to investigative action.

Follow the money forward

The goal isn't to flag individual transactions — it's to reveal the structure behind them. Trace fund flows from low-level mules toward the coordinators who control the operation.

Built for partial visibility

In practice, an analyst starts from a single alert and traces outward — they never have full blockchain visibility. This approach works with what's available, not what's ideal. Most academic approaches assume a complete graph.

BFS mirrors real investigative workflows

Start from what you know (a flagged transaction), expand forward following the money, stop at licit boundaries. This is how investigators trace funds manually — the algorithm replicates it systematically.

Excludes innocent parties by design

By expanding only through illicit nodes, the network isolates suspicious activity and excludes licit transactions entirely. Reduces noise and protects against guilt by association.

TAKEAWAY

Most AML research asks "can we classify transactions?" and stops. This thesis asks the next question: once classified, how do you show an investigator which network to examine, and in what order? Network construction bridges classification and investigative action.

NETWORKS

Network Results & Topology

The 3,254 multi-node subnetworks provide investigators with structured leads to examine. Most are small and shallow, but a meaningful subset exhibits deeper, multi-hop patterns consistent with layering behaviour described in AML research. These are potential structures of interest — flagged for investigation as suspicious networks.

Size distribution

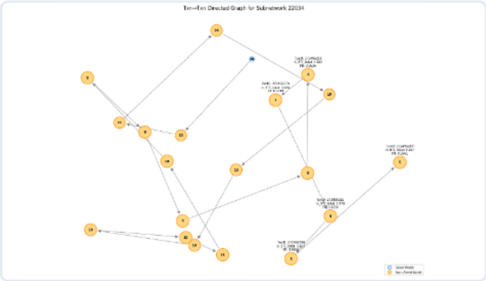
NETWORK SIZE	COUNT	%
1 node (isolated)	14,610	81.8%
2 nodes	2,053	11.5%
3–10 nodes	884	5.0%
11–100 nodes	305	1.7%
101+ nodes	**12**	**0.07%**

Depth distribution (multi-node only)

DEPTH	COUNT	%
2 (one hop)	517	54.8%
3	195	20.7%
4	103	10.9%
5+	**129**	**13.7%**

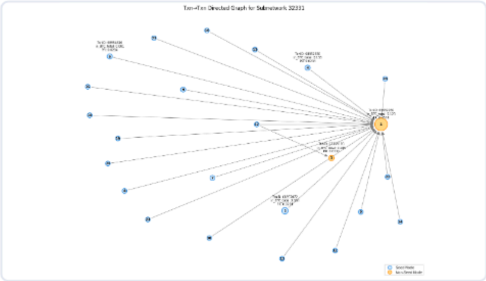
Deeper networks (5+ hops) are the most investigatively valuable — multi-layered structures where funds move through multiple intermediaries to obscure origin and ownership.

Topology patterns



Chain-like (layering)

Sequential fund flows suggesting potential layering behaviour



Star-like (hub)

May represent exchanges or services involved in illicit activity — shallow depth limits value for tracing fund flows

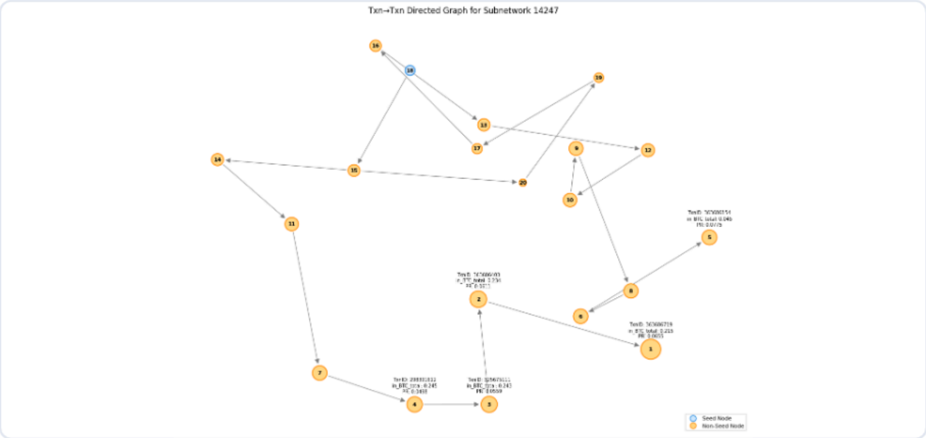
TAKEAWAY

Chain-like networks are the most valuable for investigation — their depth reveals potential layering of funds across multiple hops, exposing the structure of possible criminal operations. Star-like networks may represent exchanges or services and are less actionable due to their shallow depth. An important finding: deep chains at the transaction level often collapse into stars at the address level due to wallet reuse — investigators need both views to avoid misidentifying service nodes as criminal actors.

Example: Subnetwork 14247

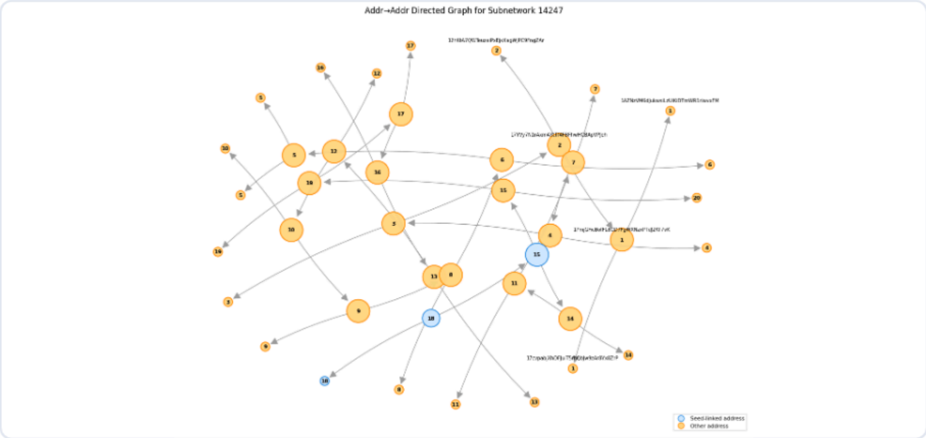
The same suspicious network viewed from two perspectives. Node size represents structural centrality (PageRank). Orange nodes are seed transactions; blue nodes are connected illicit transactions discovered through forward expansion.

Transaction-to-Transaction View



Shows how funds flow between individual transactions. Chain-like structure reveals multi-hop movement of funds — consistent with potential layering behaviour.

Address-to-Address View



Aggregates transactions to the wallet level. The chain collapses into a denser structure — address reuse and consolidation reveal entity-level relationships not visible in the transaction view.

TAKEAWAY

Investigators need both views. The transaction view shows the sequence and depth of fund flows. The address view reveals which wallets are connected and whether apparent complexity is driven by address reuse rather than distinct actors. Together, they reduce the risk of misidentifying service nodes as criminal entities.

RANKING

Why Rank Transactions?

Classification identifies *which* transactions are suspicious. Network construction reveals *how* they connect. But an investigator still needs to know *where to start*. With 3,254 multi-node networks containing thousands of flagged transactions, reviewing them equally isn't possible — ranking transforms networks into prioritised investigative roadmaps.

Investigators have limited resources

Compliance teams can't review every flagged transaction with equal depth. Without prioritisation, time is spent on low-impact nodes while central laundering pathways remain undiscovered.

Single metrics aren't enough

PageRank alone captures structural importance but ignores financial value. BTC volume alone highlights large transactions but misses structurally central nodes. No single measure captures the full picture.

Combine structure and value

The ranking algorithm integrates structural centrality, financial magnitude, connectivity, and accumulation behaviour into a single composite score — capturing both *where* a transaction sits and *how much value* flows through it.

Transparent and auditable

Every component and weight is explicitly defined. An investigator or regulator can ask "why is this transaction Rank 1?" and get a clear, defensible answer — unlike black-box scoring methods.

TAKEAWAY

Ranking bridges the gap between detection and investigation. It answers the question most AML research doesn't ask: once you've found suspicious networks, which transactions within them should an investigator examine first?

Ranking Methodology

Each transaction within a subnetwork is scored using a weighted composite of four indicators — designed from domain knowledge of how money laundering operates on the Bitcoin blockchain. The components reflect what AML investigators actually look for: structural influence, financial magnitude, source diversity, and accumulation behaviour.

STEP 1: COMPOSITE SCORE

0.60×PR + 0.30×InBTC + 0.07×InDeg + 0.03×(100−OutDeg)

Weighted sum of percentile-normalised components

→

STEP 2: COMPOSITE PERCENTILE

Score → Percentile (0–100)

Enables comparison across networks of different sizes

→

STEP 3: COMPOSITE RANK

Rank 1 → Rank N

Investigator starts at Rank 1 — the actionable output

COMPONENT	WEIGHT	WHAT IT MEASURES	AML RATIONALE
PageRank	60%	Structural centrality via recursive inbound links	Identifies who controls the flow — not just who has the most connections
Inbound BTC	30%	Total Bitcoin value received	Identifies where funds converge — aggregation and collection points
In-degree	7%	Number of distinct incoming transactions	Consolidation from multiple sources — a hallmark of layering
Inv. Out-degree	3%	Penalises high fund dispersal	Favours transactions that retain value — accumulation over distribution

All components percentile-normalised within each subnetwork — preventing high-volume service nodes (exchanges, mixers) from dominating results. Weights validated through sensitivity analysis across 1,111 subnetworks.

TAKEAWAY

This ranking methodology is where AML domain expertise meets data science. The choice of components, the decision to invert out-degree, the weighting toward structural centrality over raw volume — these reflect how laundering actually works: funds are consolidated, layered through intermediaries, and controlled by central actors. The composite score encodes that investigative logic into a transparent, auditable metric.

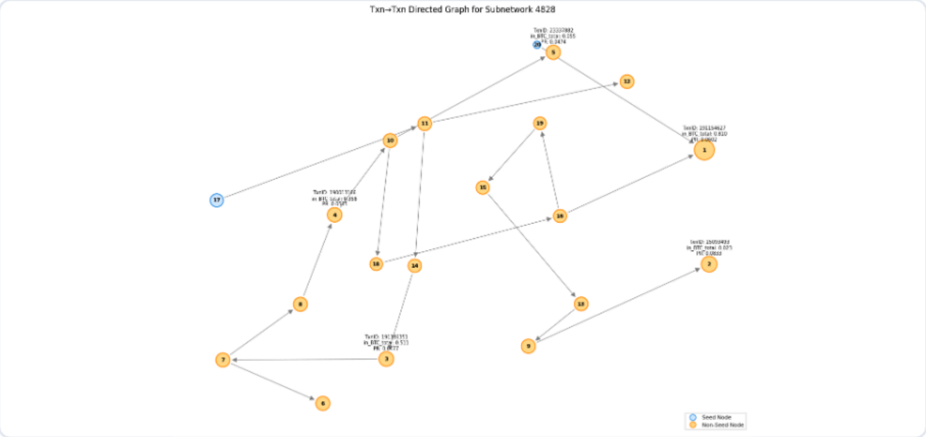
See Appendix: Weight Sensitivity & Method Comparison

RANKING

Ranked Network: Subnetwork 4828

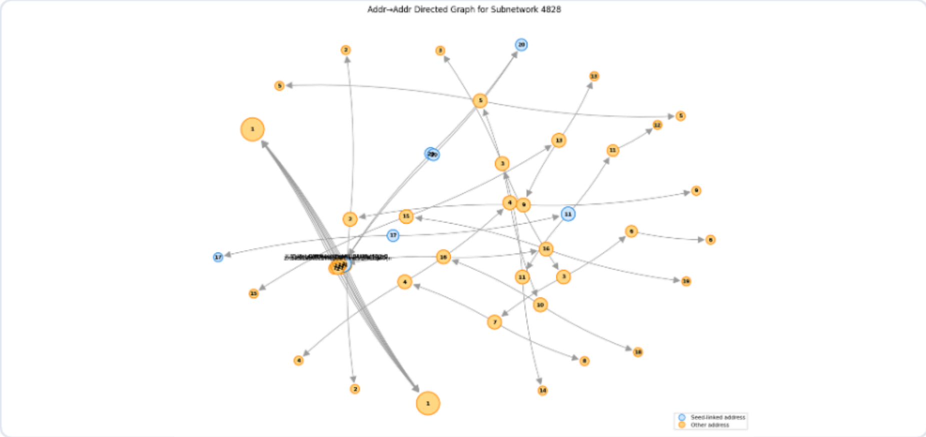
Once ranked, each transaction receives a composite score and position within its network. Node size reflects structural centrality (PageRank). Node labels show the investigative rank — Rank 1 is where the investigator starts.

Transaction-to-Transaction View



The ranked transaction flow. Investigators trace from the highest-ranked node outward — following the money through each hop.

Address-to-Address View



The same network at the wallet level. Ranking identifies which wallets are most central. Address reuse patterns help distinguish distinct actors from single entities.

TAKEAWAY

The ranking algorithm turns a network of flagged transactions into a prioritised investigative workflow — Rank 1 is where you start. But ranked data alone isn't enough. Investigators need an interactive, visual way to explore these networks. The final component brings ranking, networks, and transaction detail together in a Streamlit dashboard deployed on Google Cloud Run.

VISUALISATION

Investigator Dashboard

The final component translates all analytical outputs into an interactive, web-based dashboard built with Streamlit and deployed on Google Cloud Run. This is where classification, network construction, and ranking converge into a single investigative tool — accessible via browser with no installation required.

Dual network views

Each subnetwork is rendered in two perspectives: transaction-to-transaction (fund flows) and address-to-address (wallet relationships). Node size reflects PageRank centrality; labels show composite rank.

Ranked summary table

Accompanies each network with investigation order, transaction IDs, wallet addresses, BTC values, composite scores, and hop depth. The structured, auditable trail investigators need to build a case.

Interactive exploration

Investigators select a subnetwork and explore it interactively — tracing fund paths and examining transaction detail via the summary table. No static reports; the data is live from BigQuery.

Built for investigators, not data scientists

No coding required. No model knowledge needed. The dashboard presents ranked networks in a format compliance analysts can interpret and act on — bridging the gap between machine learning and investigative action.

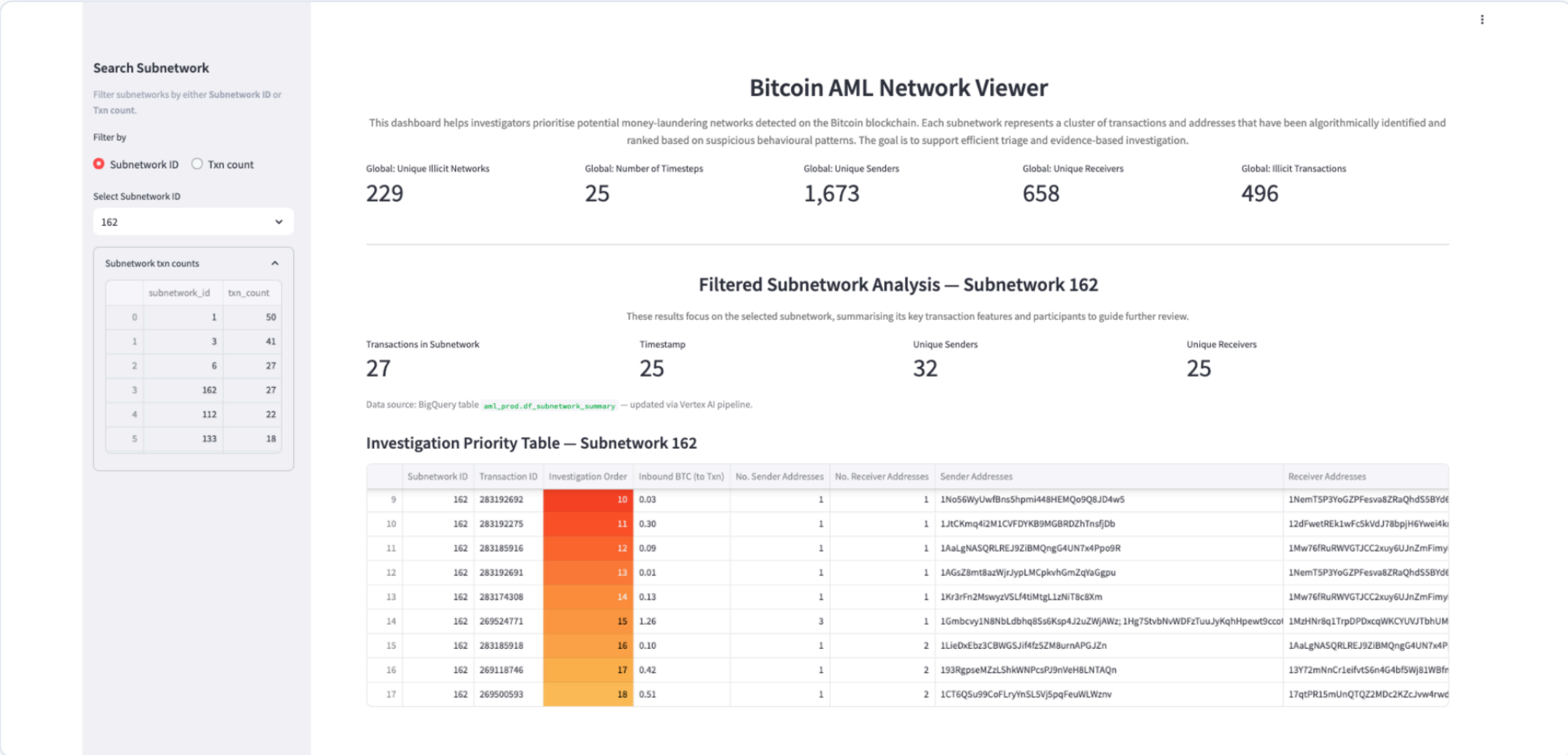
TAKEAWAY

Visualisation is the bridge between automated detection and human investigation. It converts complex analytical outputs into clear, interpretable artefacts that direct investigative attention toward key actors and relationships — enhancing explainability, supporting regulatory auditability, and ensuring the methodology remains operationally useful.

VISUALISATION

Dashboard: Overview & Investigation Priority

The landing view provides summary statistics across all subnetworks, filtered analysis for a selected subnetwork, and a ranked investigation priority table with transaction IDs, wallet addresses, BTC values, and composite scores.

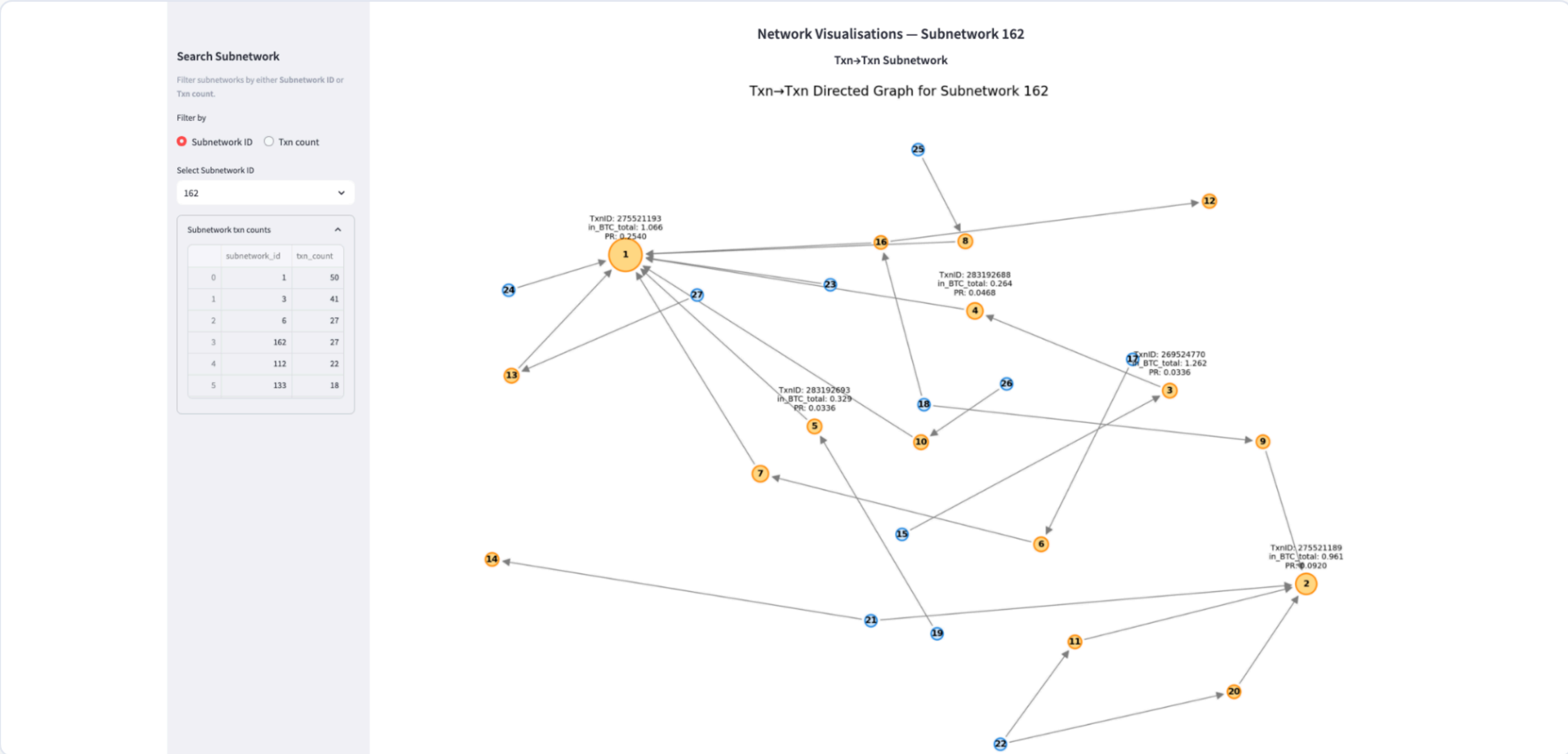


	subnetwork_id	txn_count
0	1	50
1	3	41
2	6	27
3	162	27
4	112	22
5	133	18

	Subnetwork ID	Transaction ID	Investigation Order	Inbound BTC (to Txn)	No. Sender Addresses	No. Receiver Addresses	Sender Addresses	Receiver Addresses
9	162	283192692	10	0.03	1	1	1No56WyUwfBns5hpmi448HEMQo9Q8JD4w5	1NemT5P3YoGZPFesva8ZRaQhdS5BYd
10	162	283192275	11	0.30	1	1	1JtCKmq4i2M1CVFDYKB9MGBRDZhTnsfjDb	12dFwetREk1wFc5kVdJ78bpjH6Ywei4k
11	162	283185916	12	0.09	1	1	1AaLgNASQRLREJ9ZiBMQngG4UN7x4Ppo9R	1Mw76fRuRWVGTJCC2xuy6UJnZmFimy
12	162	283192691	13	0.01	1	1	1AGsZ8mt8azWjrJypLMCpkvhGmZqYaGgpu	1NemT5P3YoGZPFesva8ZRaQhdS5BYd
13	162	283174308	14	0.13	1	1	1Kr3rFn2MswyzVSLf4tiMtgL1zNiT8c8Xm	1Mw76fRuRWVGTJCC2xuy6UJnZmFimy
14	162	269524771	15	1.26	3	1	1Gmbcvy1N8NbLdbhq8Ss6Ksp4J2uZWjAWz; 1Hg7StvbNvWDFzTuuJyKqhHpewt9cco	1MzHNr8q1TrpDPDxcqWKCYUVJTbhUM
15	162	283185918	16	0.10	1	2	1LieDxEbz3CBWGSJif4fz5ZM8urnAPGJZn	1AaLgNASQRLREJ9ZiBMQngG4UN7x4P
16	162	269118746	17	0.42	1	2	193RgpseMZzLShkWNPcsPJ9nVeH8LNTAQn	13Y72mNnCr1eifvtS6n4G4bf5Wj81WBf
17	162	269500593	18	0.51	1	2	1CT6QSu99CoFLryYnSL5Vj5pqFeuWLWznv	17qtPR15mUnQTQZ2MDc2KZcJvw4rwd

Dashboard: Transaction Network View

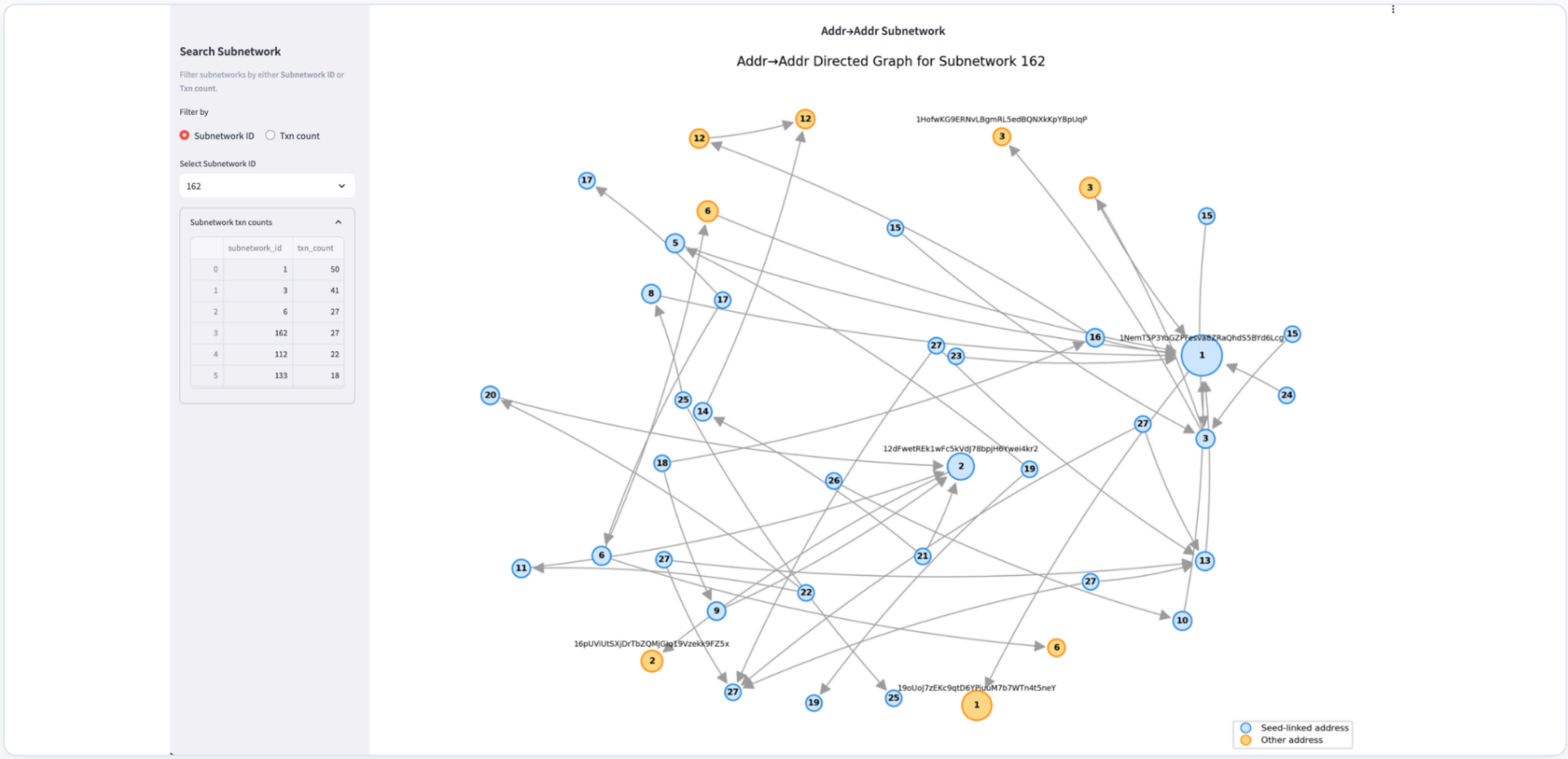
The transaction-to-transaction directed graph for Subnetwork 162. Node size reflects PageRank centrality, labels show composite rank. These rankings are also presented in the investigation priority table, giving investigators both a visual and tabular view of where to focus.



VISUALISATION

Dashboard: Address Network View

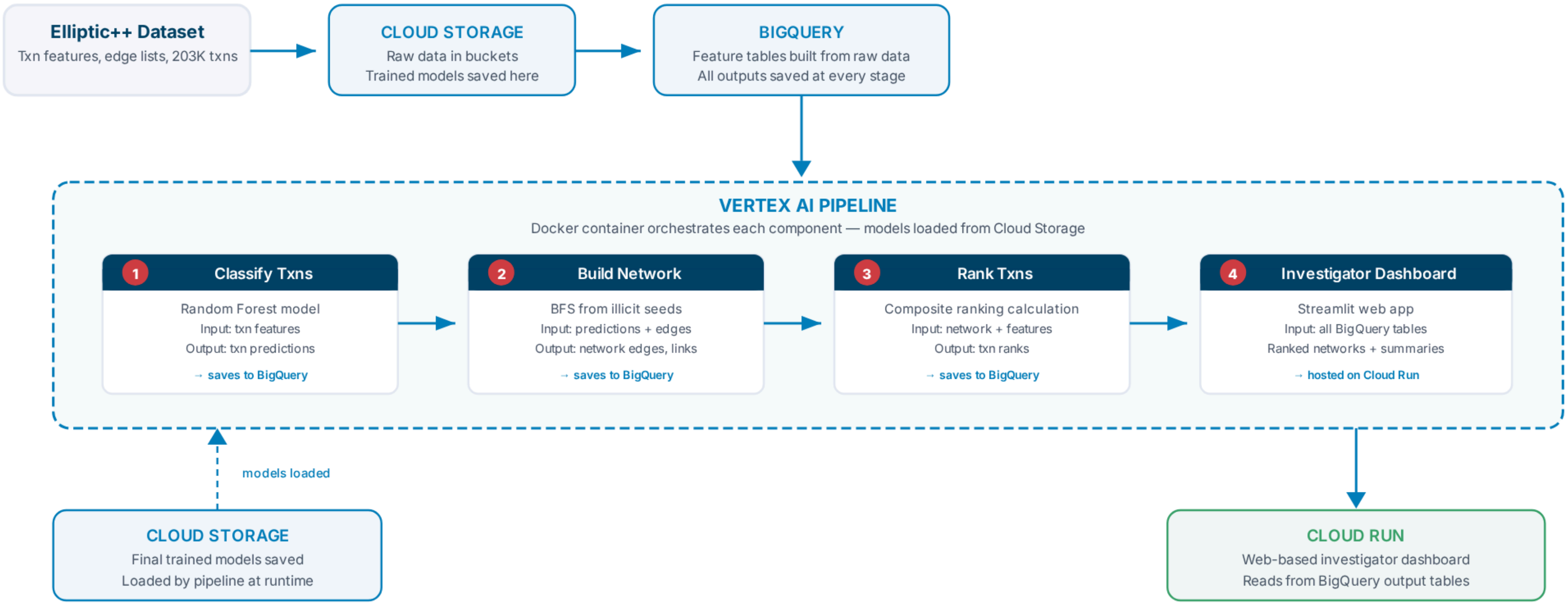
The same subnetwork at the address (wallet) level. Multiple transactions collapse into shared wallets, revealing entity-level relationships. The legend distinguishes same-linked addresses from other addresses — both views together give investigators the full picture.



DEPLOYMENT

Automating the Methodology in Google Cloud

The entire methodology — from classification to visualisation — is automated as a cloud-native pipeline in Google Cloud Vertex AI. Each component is containerised, independent, and writes to BigQuery at every stage for full auditability.



TAKEAWAY

Cloud-native, modular, and fully automated. Each component is containerised and independent — classify, trace, rank, visualise. Every stage writes to BigQuery for full auditability. This is production-grade infrastructure, not a research prototype.

Limitations & Future Work

Key Limitations

Classifier dependency

All downstream components depend on classification accuracy. False negatives fragment network paths; false positives introduce innocent transactions. At threshold 0.4: 535 missed illicit, 652 false alerts.

Dataset age & labels

The Elliptic dataset reflects 2016–2017 laundering typologies. Since then, techniques have evolved to include cross-chain swaps, privacy coins, and decentralised mixers. Ground truth labels are proprietary and cannot be independently verified.

No wallet-type metadata

Cannot distinguish exchanges, mixers, or payment processors from end-user wallets. Large legitimate service nodes can dominate network connectivity, introducing structural noise.

Batch processing only

Developed under constrained computing resources without live blockchain integration. Processing is batch-based, limiting temporal analysis and real-time monitoring capability.

Future Work

Entity-level analysis

Link transaction-level subnetworks to wallet or entity-level clusters to identify exchanges, mixers, and service providers. Integrate off-chain context for clearer actor identification.

Real-time monitoring

Incorporate live blockchain data feeds and near real-time analytics for continuous monitoring. Temporal analysis could reveal evolving or recurring laundering typologies over time.

Cross-chain generalisability

Apply the framework to other blockchains — Ethereum, cross-chain ecosystems — to assess how well the design principles of transparency, explainability, and ranking generalise beyond Bitcoin.

Weighted edges & modern data

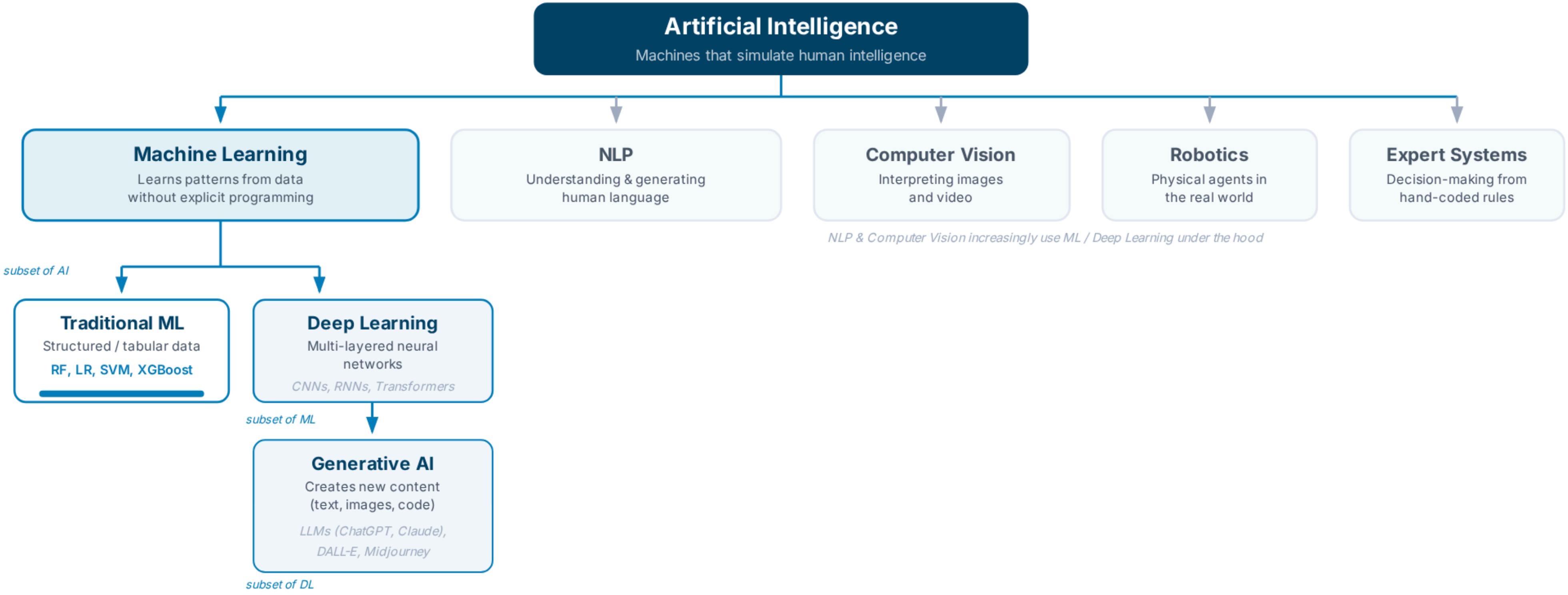
Incorporate BTC values as edge weights into network construction and PageRank calculation. Retrain on current data to account for evolved laundering techniques and domain drift.

TAKEAWAY

These limitations are acknowledged, not hidden. Every constraint was a deliberate trade-off favouring interpretability, auditability, and operational relevance — the qualities that make this framework usable in a regulated environment. The future work builds on a proven foundation.

Artificial Intelligence: A Quick Guide

Artificial Intelligence (AI) is the broad field of building machines that can perform tasks typically requiring human intelligence. The diagram below shows how the key subfields relate — each deeper level is a **subset** of the one above it.

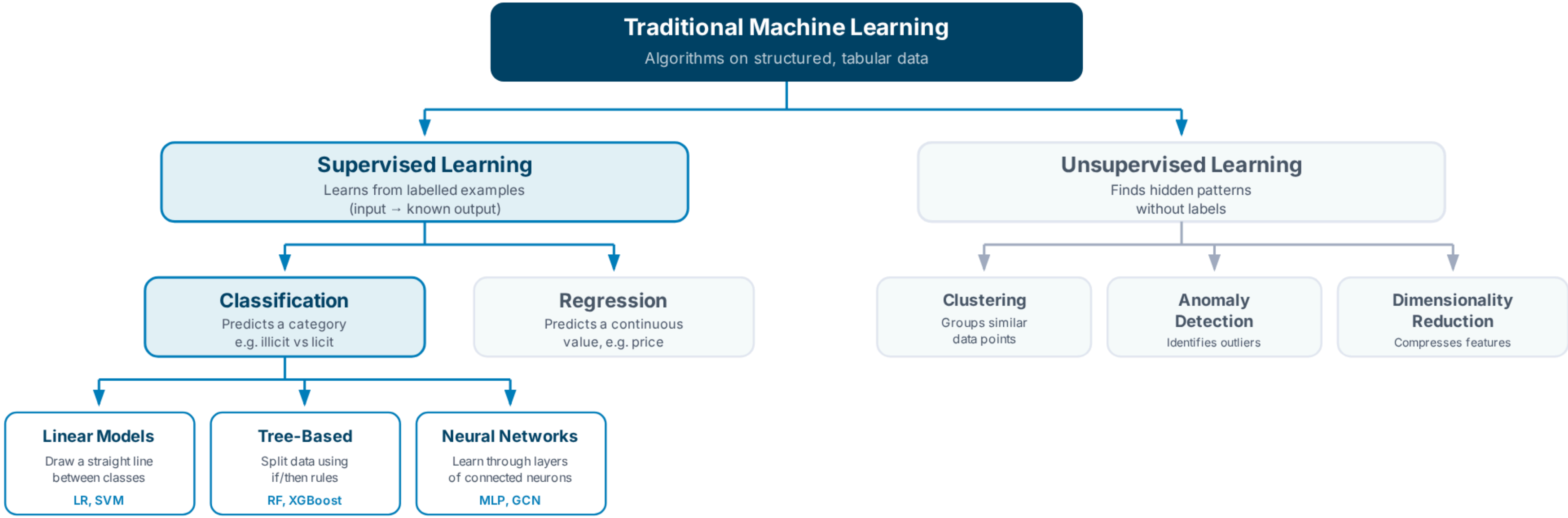


This thesis sits in **Traditional ML** (marked above) — using supervised classification on structured transaction data. This layer prioritises interpretability and auditability over the raw power of deep learning, which is critical in a regulated AML environment. The next slide breaks down the ML landscape in detail.

APPENDIX

Traditional Machine Learning: A Quick Guide

Traditional machine learning uses algorithms on structured, tabular data — the kind of data found in spreadsheets and databases. The diagram below shows how traditional ML methods are organised and where this thesis sits.



This thesis: Supervised Learning → Classification → **Logistic Regression** (baseline) & **Random Forest** (optimised). Selected for interpretability, auditability, and peer-reviewed precedent on the Elliptic dataset.

Key terms: A **label** is a known answer (e.g. "this transaction is illicit"). A **feature** is a measurable property of the data (e.g. transaction fee, volume). A **model** is an algorithm trained on labelled data to make predictions on new, unseen examples.

Classification Models Overview (1 of 2)

Classification is a form of supervised learning where a model is trained on labelled examples — transactions already identified as illicit or licit — and learns to predict the label for unseen data. The models below represent different approaches to learning that decision boundary, each with distinct trade-offs in accuracy, interpretability, and scalability.

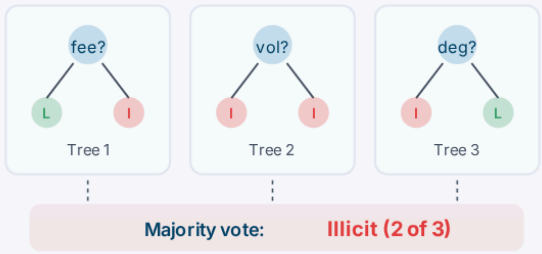
Logistic Regression

Definition: A linear model that multiplies each feature by a learned weight, sums them, and passes the result through a sigmoid function to produce a probability between 0 and 1.

Pros: Every weight is directly interpretable — you can see exactly how much each feature contributes to the prediction. Fast to train, simple to implement, and ideal for regulatory audit where explainability is mandatory.

Cons: Can only learn linear (straight-line) decision boundaries. Struggles when the signal is embedded in complex, non-linear feature interactions — as demonstrated in this thesis where precision collapsed to 46% after tuning.

Random Forest

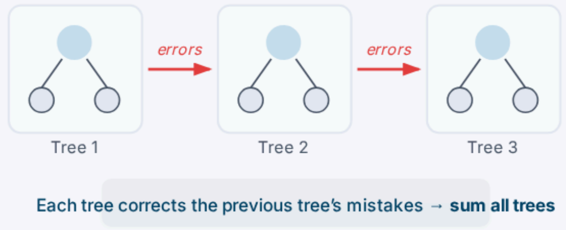


Definition: An ensemble model that builds hundreds of independent decision trees, each trained on a random subset of data and features. Each tree votes on the outcome; the majority wins.

Pros: Captures non-linear patterns by combining many simple rules. Feature importance scores show what drives predictions. Naturally resistant to overfitting due to averaging across diverse trees. Proven top performer on the Elliptic dataset.

Cons: Less interpretable than LR at the individual prediction level — you can see which features matter overall, but tracing a single prediction through hundreds of trees is harder. Can be slower to train with very large feature sets.

XGBoost / Gradient Boosting



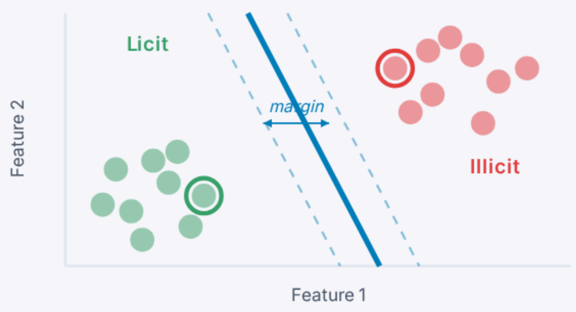
Definition: A boosting model that builds trees sequentially — each new tree specifically targets the errors the previous trees got wrong. The final prediction is the weighted sum of all trees.

Pros: Often achieves state-of-the-art accuracy on tabular data. Handles missing values and class imbalance natively. Highly tuneable with built-in regularisation.

Cons: Sequential construction makes individual predictions harder to trace and audit. More prone to overfitting without careful tuning. Each tree depends on the previous, making the overall model harder to explain than Random Forest.

Classification Models Overview (2 of 2)

Support Vector Machine

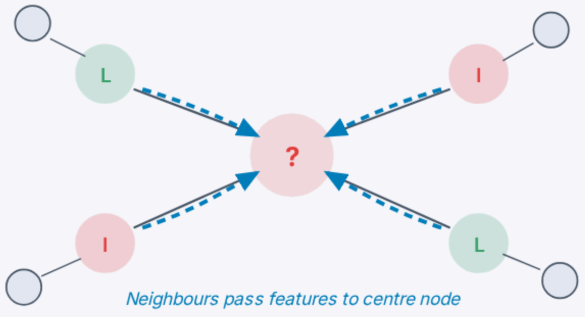


Definition: A model that finds the boundary (hyperplane) which maximises the gap between two classes. The points closest to the boundary are called "support vectors" and define the decision surface.

Pros: Effective in high-dimensional spaces. Strong theoretical foundation with clear margin optimisation. Can handle non-linear boundaries via kernel functions.

Cons: Scales poorly with large datasets (200K+ transactions make training computationally expensive). No published precedent on the Elliptic dataset. Kernel choice adds complexity and reduces interpretability.

Graph Convolutional Network

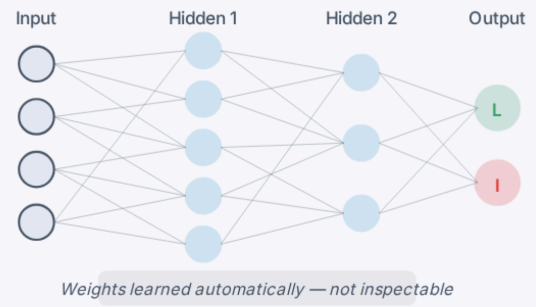


Definition: A neural network designed for graph-structured data. Each node collects and aggregates feature information from its neighbours through "message passing" — after multiple rounds, a node's representation encodes its local graph structure.

Pros: Natively exploits graph structure — learns from both transaction features and network topology simultaneously. Can capture complex relational patterns that tabular models miss.

Cons: Black-box — cannot explain why a specific transaction was flagged, which fails regulatory scrutiny. Requires the full graph at inference time. Weber et al. tested GCN on Elliptic and it underperformed Random Forest.

Multi-Layer Perceptron (MLP)



Definition: A neural network that passes input through layers of interconnected artificial neurons. Each connection has a learned weight, and multiple hidden layers enable the model to learn complex, non-linear patterns.

Pros: Universal approximator — can learn arbitrarily complex decision boundaries given enough data. Flexible architecture adaptable to many problem types.

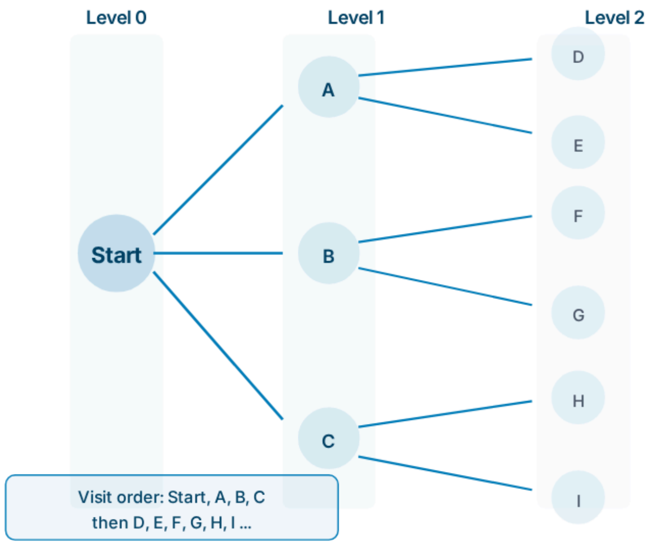
Cons: Thousands of internal weights with no interpretable meaning — a black box that cannot satisfy regulatory explainability requirements. Requires large datasets and careful regularisation to avoid overfitting. No clear advantage over RF on structured tabular data like Elliptic.

APPENDIX

What is Breadth-First Search (BFS)?

BFS is a graph traversal algorithm that explores all neighbours at the current depth before moving to the next level. In this thesis, it is used to trace fund flows outward from a seed transaction — one hop at a time.

How BFS expands level by level



How it works

- **Start** from a root node (in this thesis: a flagged illicit transaction)
- **Visit all direct neighbours** first (Level 1) before moving to their neighbours (Level 2)
- **Track depth** — each level = one "hop" from the starting point
- **Never revisit** a node already seen — prevents infinite loops
- **Deterministic** — same input always produces the same output

How this thesis adapts it

- **Forward only** — follows the direction of fund flows (downstream)
- **Illicit to Illicit only** — expands only through flagged transactions
- **Stops at licit boundaries** — excludes innocent parties
- **No depth limit** — follows the trail as far as it goes

APPENDIX

Ranking: Weight Sensitivity & Method Comparison

Multiple weighting schemes were tested across 1,111 subnetworks (≥5 transactions). For each configuration, the top 20% of ranked nodes were examined to measure how much BTC value and PageRank influence they captured.

Weight configurations tested

CONFIG	PR	VAL	IN	OUT	PR PCT	BTC %
PR extreme	0.80	0.15	0.03	0.02	92.3	28.1%
PR heavy	0.70	0.20	0.07	0.03	92.3	29.3%
Selected	**0.60**	**0.30**	**0.07**	**0.03**	**91.7**	**32.1%**
Balanced	0.50	0.40	0.07	0.03	91.1	35.0%
Equal	0.25	0.25	0.25	0.25	90.7	32.4%
Deg heavy	0.40	0.30	0.20	0.10	91.2	33.2%
Val mid	0.40	0.50	0.06	0.04	80.0	43.5%
Val heavy	0.20	0.70	0.06	0.04	56.3	51.1%
Val extreme	0.10	0.80	0.05	0.05	43.8	52.5%

The selected weighting maintains high structural alignment (PR ≈91.7) while capturing meaningful financial concentration (32% of subnetwork BTC). Value-heavy configs collapse PR percentile — structurally unimportant nodes ranked highly just because they handle large volumes.

Other methods evaluated

METHOD	LIMITATION
HITS (hub/authority)	Unstable in small directed graphs
Betweenness	Prioritises transient intermediaries over value controllers
Eigenvector / Katz	Overemphasise dense clusters
Degree centrality	Counts connections, not influence
Coreness	Ignores direction and value

Spearman correlation analysis across 1,111 subnetworks confirmed the composite captures a unique signal — correlating strongly with PageRank (ρ=0.88), harmonic (ρ=0.86), and Katz (ρ=0.85) centrality, but diverging from degree and coreness measures. This confirms the composite integrates structural influence while adding financial context no single metric provides.